

## **EXPERIMENT- 6**

1.

# Dataset

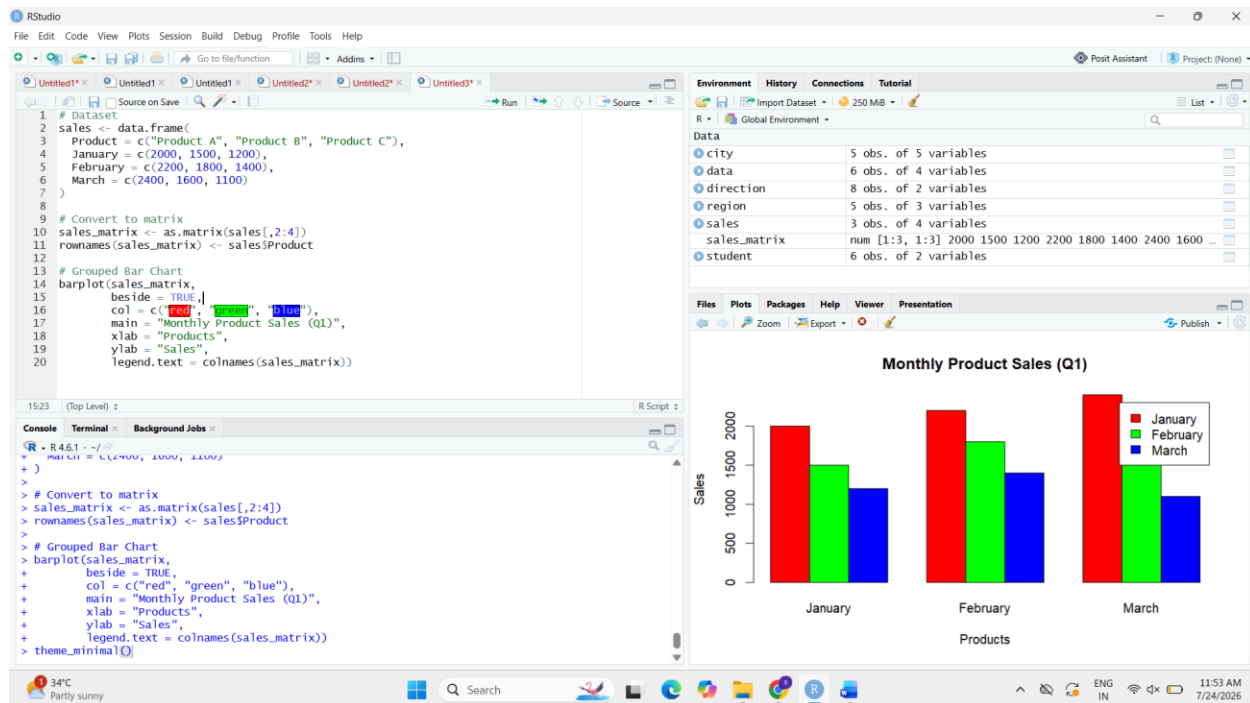
```
sales <- data.frame(  
  Product = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)
```

# Convert to matrix

```
sales_matrix <- as.matrix(sales[,2:4])  
rownames(sales_matrix) <- sales$Product
```

# Grouped Bar Chart

```
barplot(sales_matrix,  
  beside = TRUE,  
  col = c("red", "green", "blue"),  
  main = "Monthly Product Sales (Q1)",  
  xlab = "Products",  
  ylab = "Sales",  
  legend.text = colnames(sales_matrix))
```



2.

# Dataset

sales <- data.frame(

Month = c("January", "February", "March"),

ProductA = c(2000, 2200, 2400),

ProductB = c(1500, 1800, 1600),

ProductC = c(1200, 1400, 1100)

)

# Install package (only first time)

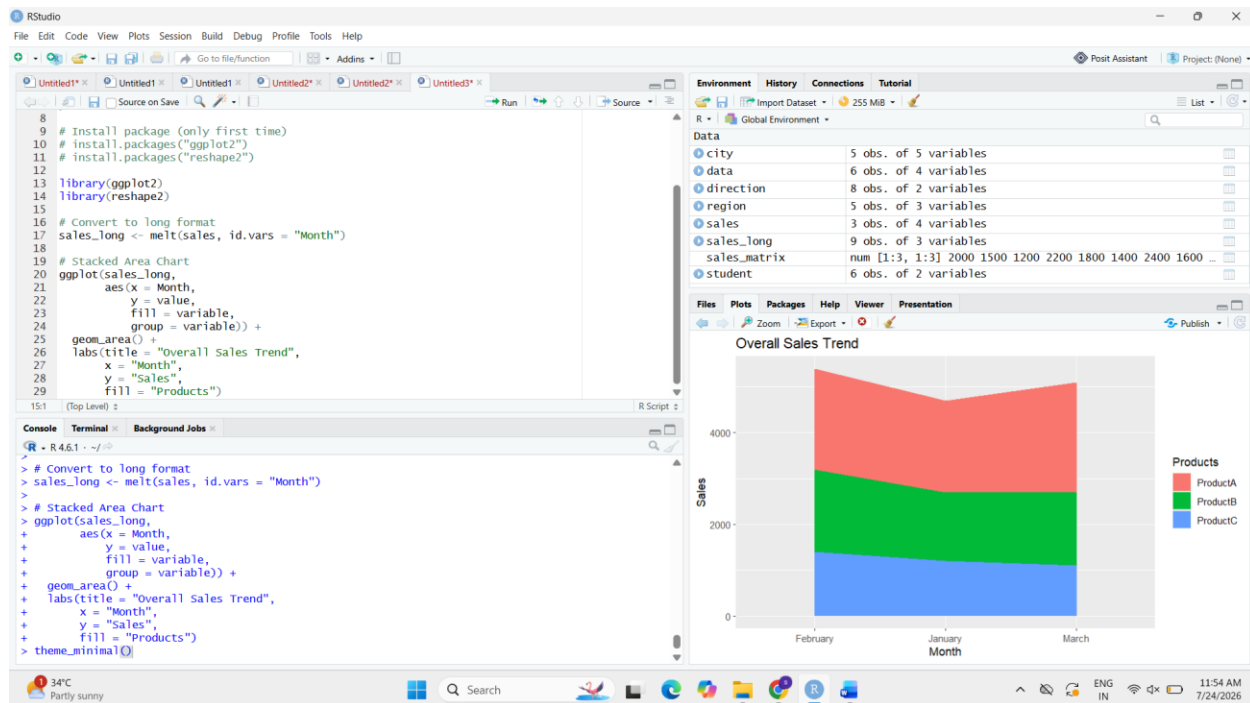
# install.packages("ggplot2")

# install.packages("reshape2")

```
library(ggplot2)
library(reshape2)

# Convert to long format
sales_long <- melt(sales, id.vars = "Month")

# Stacked Area Chart
ggplot(sales_long,
       aes(x = Month,
           y = value,
           fill = variable,
           group = variable)) +
  geom_area() +
  labs(title = "Overall Sales Trend",
       x = "Month",
       y = "Sales",
       fill = "Products")
```



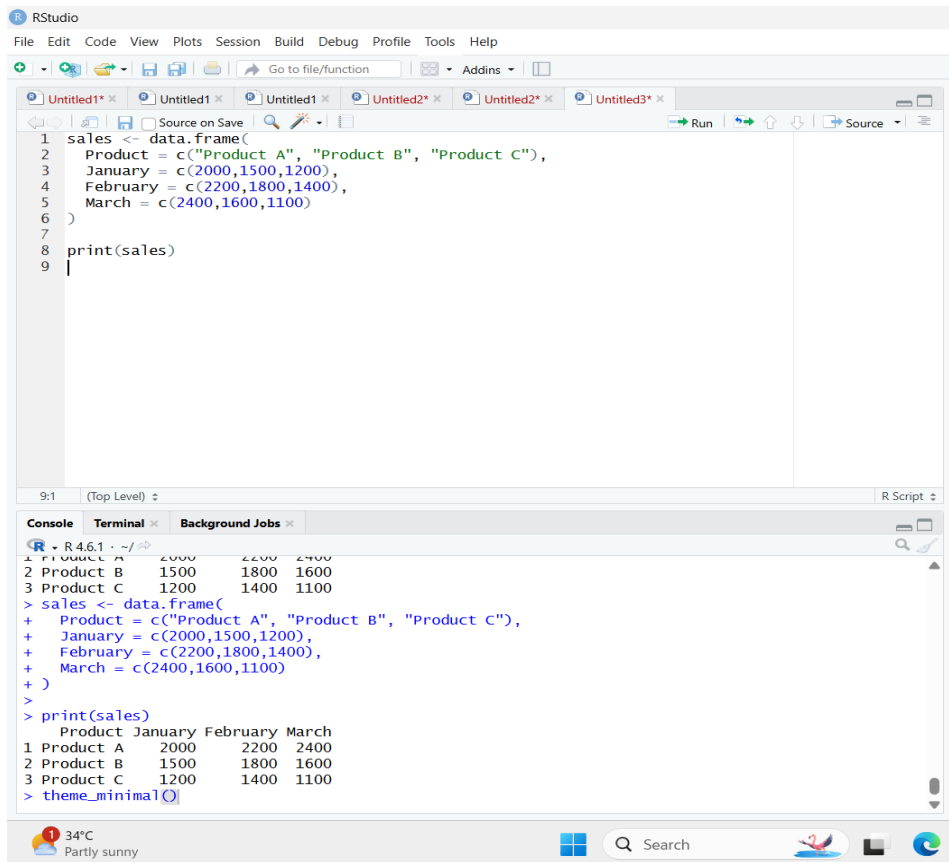
3.

```

sales <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

```

```
print(sales)
```



4.

